

Guest Editorial: Materials & Devices for Advanced Flexible Sensors

Sensors serve as very important parts of electronic systems, which help electronics to acquire information from environment or human. As rigid electronics are not suitable for use on curved surfaces such as human body, advanced flexible sensors become very essential for these kinds of applications. To realize high quality flexible sensors that have similar or even advanced performance compared to rigid sensors, various advanced materials, design strategies, and device formats have been developed, which are what the papers in this special section will discuss about. There are four review articles to summarize recent advances in skin electronics, touch sensors for flexible display, near-infrared spectroscopy (NIRS) and organic electrochemical transistors (OECT), respectively. Three research article introduces new methods in flexible pressure sensing array, ammonia sensors and charge plasma junctionless tunnel field effect transistor (CP JLTFET), respectively.

In the first review article “Recent Advances and Design Strategies Towards Wearable Near-infrared Spectroscopy”, S. Liu et al. [A1] systematically introduced the principle and design strategies for NIRS in wearable applications such as physiological monitoring of oxygenation status, heart rate, blood pressure, etc. Examples of advanced wearable NIRS devices based on design strategies including modular device design, flexible hybrid electronics (FHE) technology with single or multiple source-detector distances, and materials innovations are illustrated in detail. The future development direction of wearable NIRS was also discussed, which might become an important part in digital healthcare technologies.

Precise physical information sensing such as touching is a key part in interactive human-machine interfaces, for example, touch screen is an essential input for flexible display. In review article “Recent Advances in Touch Sensors for Flexible Displays”, C. Ouyang et al. [A2] summarized various mechanics and structures of sensors for flexible touch screen and discussed their benefits and drawbacks. Instead of out-cell structure, on-cell, in-cell and hybrid-cell style are more used in integrated touch. Mechanisms including surface capacitive, self-capacitive and mutual capacitive are introduced and compared. Moreover, they also summarized several advanced materials for flexible touch screen sensors.

Particularly using capacitive sensing mechanism, graphene oxide (GO) could be a great material option for enhancing performance. In the paper “Droplet-Dispensed Graphene Oxide as Capacitive Sensing Elements for Flexible Pressure-Pulse Sensing Array”, K.W. Kong et al. [A3] proposed a novel method of fabricating pressure sensing array by using GO-dispersed droplets. With lower than 70 μL of GO aqueous

dispersion per droplet, it reached a sensitivity of 10^{-3} kPa^{-1} . This method is promising in rapidly fabricating batches of sensors with high repeatability, which has been successfully demonstrated in pulse sensing on human body.

When it comes to the sensing on the body, the format of soft sensors or systems should be soft, thin and could conformally attach to the skin, which is also called skin electronics. In “Recent Advances in Materials, Designs and Applications of Skin Electronics”, K. Yao et al. [A4] reviewed the advances in the development of skin electronics in the past decade, and highlighted them in three categories. For materials innovation, devices using functional materials, conductive polymers, carbon nanomaterials, hydrogels were introduced. Mechanics design and comfortability design strategies to make skin electronics stretchable and comfortable for wearing were also summarized. Lastly, skin electronics can be used in various applications, including physiological and healthcare sensing, biomedical imaging, and human-machine interface (HMI), etc.

As one of the most promising sensing technique, organic electrochemical transistors (OECT) show excellent performance in sensitivity, biocompatibility, low driving voltage, etc. In “Integrated Sensing Arrays Based on Organic Electrochemical Transistors” [A5], from its mechanism, typical organic mixed ionic-electronic (semi)conductors (OMIEC) such as PEDOT: PSS, p(a2T-TT) and p(g2T-TT) and their performances, to novel applications for various bioelectrical sensing, J. Wen et al. systematically introduced OECT. They classified OECT sensing arrays into two major types, passive and active arrays, listed examples and compared their pros and cons. Perspectives for the future development of OECT in aspects of the stability, effective bandwidth, integration density, and power dissipation were also discussed in this review.

Another kind of organic transistor, organic field-effect transistors (OFET) are normally based on uniform organic semiconductor (OSC) films. However, it was found that by utilizing nanofiber-textured OSC films, OFET-based gas sensor presented much more outstanding performance of response to ammonia. In research article “Nanofiber-Textured Organic Semiconductor Films for Field-Effect Ammonia Sensors”, Y. Tang et al. [A6] demonstrated this strategy and found that the resulting OFET achieved a remarkably high ammonia sensitivity of 13676%/ppm (current) and 457%/ppm (turn-on voltage), respectively.

In the last paper “Design and Parametric Analysis of Charge Plasma Junctionless TFET for biosensor applications”, D. Manasawi and S. R. Karumuri [A7] proposed a

creative design of charge plasma junctionless tunnel field effect transistor (CP JLTFET) which obtained an improved ON current and surface potential for various nano-cavity regions. With the length of cavity varying from 8 to 10 nm, biological molecules could be immobilized on the tunneling junction within the device. Analysis results showed that charge carriers recombination rate increased, which contributed to the enhancement of the performance. This CP JLTFET might be useful for detection of biological molecules both neutral and charged.

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APPENDIX: RELATED ARTICLES

- [A1] S. Liu, B. Xue, W. Yan, A. Y. Rwei, and C. Wu, "Recent advances and design strategies towards wearable near-infrared spectroscopy," *IEEE Open J. Nanotechnol.*, early access, Dec. 05, 2022, doi: 10.1109/OJNANO.2022.3226603.
- [A2] C. Ouyang, D. Liu, K. He, and J. Kang, "Recent advances in touch sensors for flexible displays," *IEEE Open J. Nanotechnol.*, early access, Nov. 25, 2022, doi: 10.1109/OJNANO.2022.3224757.
- [A3] K. W. Kong et al., "Droplet-dispensed graphene oxide as capacitive sensing elements for flexible pressure-pulse sensing array," *IEEE Open J. Nanotechnol.*, early access, Nov. 21, 2022, doi: 10.1109/OJNANO.2022.3223712.
- [A4] K. Yao, Y. Yang, P. Wu, G. Zhao, L. Wang, and X. Yu, "Recent advances in materials, designs and applications of skin electronics," *IEEE Open J. Nanotechnol.*, early access, Nov. 02, 2022, doi: 10.1109/ojnano.2022.3218960.
- [A5] J. Wen, J. Xu, W. Huang, C. Chen, L. Bai, and Y. Cheng, "Integrated sensing arrays based on organic electrochemical transistors," *IEEE Open J. Nanotechnol.*, vol. 3, pp. 101–115, 2022, doi: 10.1109/OJNANO.2022.3215135.
- [A6] Y. Tang et al., "Nanofiber-textured organic semiconductor films for field-effect ammonia sensors," *IEEE Open J. Nanotechnol.*, vol. 3, pp. 116–123, 2022, doi: 10.1109/ojnano.2022.3217255.
- [A7] D. Manasawi and S. R. Karumuri, "Design and parametric analysis of charge plasma junctionless TFET for biosensor applications," *IEEE Open J. Nanotechnol.*, early access, Nov. 24, 2022, doi: 10.1109/OJNANO.2022.3224462.